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Report on the PhD thesis of Samuel Aychet-Claisse: “Digital and analog quantum algorithms for the nuclear many-body problem”, Université Paris-Saclay (CEA Saclay and IJCLab Orsay), supervised by Denis Lacroix and co-supervised by Vittorio Somà

Summary of the thesis. The thesis explores how hybrid quantum–classical variational algorithms can be applied to the nuclear many-body problem, on both digital and analog quantum hardware. It is organized in six chapters (117 pages of main text, six appendices, 109 references). Chapters 1 and 2 introduce the second-quantized many-body formalism, an overview of nuclear theory including one-body Green’s functions, the two test models used throughout (the pairing Hamiltonian and a leading-order pionless-EFT model of the deuteron), and the elements of digital quantum computing, including the Jordan–Wigner encoding and the grouping of Pauli strings for measurement. Chapter 3 defines a generalized ADAPT-VQE scheme and four concrete variants (ADAPT-St, ADAPT-Min, ADAPT-Min2, ADAPT-Fix) that differ in the operator pool and in the selection criterion; they are benchmarked on the pairing model with $A = 8$ particles in $D = 8$ levels against exact results and against projected BCS ansätze. Chapter 4 uses the resulting ground states as input to a quantum-subspace-expansion procedure that reconstructs the one-body Green’s function from its Lehmann representation, with a detailed analysis of the number of measurement bases and shots required. Chapters 5 and 6 turn to analog quantum computing with neutral atoms, with a specific focus on the Ruby machine delivered by Pasqal to the TGCC: the obstacles raised by an interaction that cannot be switched off, by global pulses and by the geometry of the register are identified, and two strategies are developed, one for a future XY-mode device with local detunings, and one, based on an alternative fermion-to-qubit encoding, that is compatible with the present hardware and results in a complete protocol for the ground-state energy of a three-level deuteron model on two atoms.

Scientific merits. The thesis is well structured and written with unusual pedagogical care: every method is derived from first principles, with the technical steps relegated to appendices, so that the document can be read by a nuclear physicist with no prior exposure to quantum computing. The scientific contributions are concrete. In Chapter 3, the observation that the energy curve along any operator of the pool has the closed form $C_0 + C_1 \cos(\theta + \gamma_1) + C_2 \cos(2\theta + \gamma_2)$ leads to the ADAPT-Min criterion, which selects the operator with the lowest attainable energy from five evaluations; the comparison of the four variants is careful and shows, among other things, that the necessarily non-unitary-rotation operators of Gard et al. can trap the adaptive procedure in a two-step loop, and that the re-optimization step is decisive below one per cent accuracy. The demonstration that 20 iterations suffice to beat the variation-after-projection BCS ansatz across the normal-to-superfluid transition is a useful benchmark for the community. Chapter 4 is the most mature part of the thesis and has been published in Physical Review C (2026). The exploitation of the block structure of the pairing Hamiltonian to reduce the $4\Omega^2$ matrix elements to $4D$, the explicit Pauli decomposition of the kernels $a_i a_j^\dagger$, the reduction of the number of measurement bases to $2D + 1$, the quantification of the Green’s function through the moments of the spectral strength, the access to odd- A energies as a by-product, and the numerical estimate that about 10^4 shots per basis are needed for one per cent accuracy, all constitute a genuine resource analysis of the kind that the field needs. The analog part is, to my knowledge, one of the first attempts to run a nuclear-physics variational calculation on a real neutral-atom machine, and the candidate has clearly engaged with the hardware at a level that is rare in a theory thesis. The characterization of the observables that can be measured with global



basis rotations only, the choice of an encoding Λ that maps the three-level deuteron into the symmetric two-qubit subspace and its Hamiltonian into a measurable operator, the choice of $\theta = \pm\pi/4$ to minimize the pulse area of the basis rotations, and the derivative condition (6.14)–(6.16) for initial detunings are all original and clever. The candidate is also refreshingly honest about limitations: the absence of quantum advantage in the Hamming-weight-1 sector, the failure to reach the pairing ground state for $2 \leq K \leq N-2$, and the 8–9 per cent energy error caused by the interaction during the basis rotations are stated plainly rather than hidden.

Conclusion. This thesis presents a coherent, carefully executed and clearly written body of work at the interface of nuclear many-body theory and quantum computing. The candidate has mastered both fields, has produced original algorithmic developments and a quantitative resource analysis that has already led to a refereed publication, and has carried a nuclear-physics application all the way to a hardware-ready protocol on a real analog quantum computer. The work fully satisfies the requirements for a doctoral degree. I therefore give a very favorable opinion and recommend without reservation that Samuel Aychet-Claisse be allowed to defend his thesis.

Sincerely,

A handwritten signature in black ink, reading "Morten Hjorth-Jensen". The script is cursive and fluid, with the first name "Morten" and last name "Hjorth-Jensen" clearly distinguishable.

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